Isolating Features via Multi-Contrastive Projection

Abstract

We analyze the problem of recovering a single feature direction from contrastive activation differences against multiple reference concepts. Standard contrastive activation addition (CAA) produces a difference vector $\mathbf{f}_1 - \mathbf{f}_j$ that captures the axis of variation between two concepts; this is effective for steering along that specific axis but conflates the target feature with the reference. Here we ask what can be gained by using *many distinct* contrast concepts for the same target. We argue that both the mean estimator and the leading singular vector of the stacked difference matrix recover the target feature with error $O(1/\sqrt{k})$, where k is the number of contrast concepts, and that these estimators are equivalent to leading order. When they diverge in practice, this serves as a diagnostic for assumption violations. We also observe that access to all pairwise contrasts can improve on the $1/\sqrt{k}$ rate.

1 Setup and Motivation

Let $\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_k \in \mathbb{R}^d$ be feature direction vectors for k concepts (e.g., languages, sentiment polarities, or stylistic attributes) in the residual stream of a transformer. We wish to isolate $\mathbf{f}_1$ using contrastive activation pairs.

For each pair of concepts $(1, j)$, we construct a set of matched prompt pairs that differ only in concept 1 versus concept j. Each prompt pair yields a difference vector $\mathbf{a}^{(i)}_{\text{pos}} - \mathbf{a}^{(i)}_{\text{neg}} \approx \mathbf{f}_1 - \mathbf{f}_j + \boldsymbol{\eta}^{(i)}$, where $\boldsymbol{\eta}^{(i)}$ is a per-sample noise term. Across n_j pairs for contrast j, these give a collection of noisy observations of $\mathbf{f}_1 - \mathbf{f}_j$.

Goal. Recover $\mathbf{f}_1$ (up to scale) from the full collection of difference vectors across all $k-1$ concept contrasts.

Why isolate $\mathbf{f}_1$? In standard CAA, the difference $\mathbf{f}_1 - \mathbf{f}_j$ is exactly what one wants: it captures the axis of variation between concepts 1 and j, and steering along it is effective. But this direction is specific to the pair $(1, j)$—it conflates the target feature $\mathbf{f}_1$ with the reference $\mathbf{f}_j$. Isolating $\mathbf{f}_1$ itself has several potential advantages:

- **Generalization.** A direction closer to $\mathbf{f}_1$ should generalize better to steering from held-out concepts $\mathbf{f}_l$ not seen during construction.
- **Validation.** If a hypothesized feature direction exists as a stable linear feature at a given layer and context, the multi-contrast procedure should recover a consistent direction. Failure to do so (e.g., high σ_2/σ_1) is informative.
- **Computational equivalence.** If one wants to steer from each of $\mathbf{f}_2, \ldots, \mathbf{f}_k$ to $\mathbf{f}_1$, computing $k-1$ separate steering vectors requires the same contrastive datasets as our method. Averaging them yields a single direction that, empirically, generalizes better to new concepts than any individual $\mathbf{f}_1 - \mathbf{f}_j$.

2 Assumptions

Assumption 1 (Near-orthogonality). $|\langle \mathbf{f}_i, \mathbf{f}_j \rangle| \leq \epsilon \|\mathbf{f}_i\| \|\mathbf{f}_j\|$ *for all* $i \neq j$*, with* ϵ *small.*

Assumption 2 (Comparable norms)**.** *There exist constants $0 < c \leq C$ such that $c \leq \|\mathbf{f}_i\| \leq C$ for all i.*

Assumption 3 (Clean contrastive pairs)**.** *For prompt pairs matched on everything except concept i vs. j:*

$$\mathbb{E}[\mathbf{a}_{\text{pos}} - \mathbf{a}_{\text{neg}}] = \mathbf{f}_i - \mathbf{f}_j + \boldsymbol{\eta}, \quad \|\boldsymbol{\eta}\| \ll \|\mathbf{f}_i - \mathbf{f}_j\|.$$

This is the standard CAA assumption: well-constructed contrastive pairs ensure confounding features cancel in expectation, leaving only the target contrast plus a small residual. It is a condition on the dataset, not the model.

3 Why a Single Contrast Does Not Suffice

With a single contrast ($k = 2$), we obtain vectors that approximate $\mathbf{f}_1 - \mathbf{f}_2$. This is useful for steering between concepts 1 and 2, but no operation on $\mathbf{f}_1 - \mathbf{f}_2$ alone can recover $\mathbf{f}_1$. Even if a second contrast $\mathbf{f}_1 - \mathbf{f}_3$ is available, projecting one onto the other gives:

$$\mathbf{p} = \frac{\langle \mathbf{f}_1 - \mathbf{f}_3, \mathbf{f}_1 - \mathbf{f}_2 \rangle}{\|\mathbf{f}_1 - \mathbf{f}_2\|^2}(\mathbf{f}_1 - \mathbf{f}_2) = \alpha(\mathbf{f}_1 - \mathbf{f}_2) + O(\epsilon),$$

where $\alpha = \|\mathbf{f}_1\|^2/(\|\mathbf{f}_1\|^2 + \|\mathbf{f}_2\|^2)$ under Assumption 1. The result is a rescaled version of $\mathbf{f}_1 - \mathbf{f}_2$—the projection merely reweights the mixture without separating $\mathbf{f}_1$ from $\mathbf{f}_2$.

4 The Mean Estimator

The simplest multi-contrast estimator averages the per-concept mean difference vectors. Let $\bar{\mathbf{v}}_{1j} = \frac{1}{n_j}\sum_{i=1}^{n_j}(\mathbf{a}_{\text{pos}}^{(i)} - \mathbf{a}_{\text{neg}}^{(i)})$ be the mean difference for contrast j. The mean estimator is:

$$\bar{\mathbf{v}} = \frac{1}{k-1}\sum_{j=2}^{k}\bar{\mathbf{v}}_{1j} \approx \mathbf{f}_1 - \underbrace{\frac{1}{k-1}\sum_{j=2}^{k}\mathbf{f}_j}_{\bar{\mathbf{f}}_{-1}}. \tag{1}$$

Each contrast contributes the full signal $\mathbf{f}_1$ (these add coherently across contrasts) plus a different reference feature $\mathbf{f}_j$. Under Assumption 1, the reference features are nearly orthogonal, so their sum grows as $\sqrt{k}$ rather than k: specifically, $\|\sum_{j=2}^{k}\mathbf{f}_j\|^2 = \sum_{j=2}^{k}\|\mathbf{f}_j\|^2 + \sum_{j\neq l}\langle \mathbf{f}_j, \mathbf{f}_l\rangle$. The cross terms are each bounded by ϵC^2, and there are $O(k^2)$ of them, giving:

$$\left\|\sum_{j=2}^{k}\mathbf{f}_j\right\|^2 = \sum_{j=2}^{k}\|\mathbf{f}_j\|^2 + O(\epsilon C^2 k^2) \approx (k-1)C^2.$$

Thus $\|\bar{\mathbf{f}}_{-1}\| \approx C/\sqrt{k-1}$.

Proposition 1 (Mean estimator error)**.** *Under Assumptions 1 and 2:*

$$\bar{\mathbf{v}} = \mathbf{f}_1 + O\left(\frac{C}{\sqrt{k-1}}\right).$$

The error is $O(1/\sqrt{k})$: the signal-to-noise ratio improves as $\sqrt{k}$ because we accumulate $k-1$ coherent copies of $\mathbf{f}_1$ against reference features whose norm grows only as $\sqrt{k-1}$.

Remark 1 (Diversity of contrasts)**.** *The $1/\sqrt{k}$ rate relies on the reference features adding incoherently. This follows from Assumption 1: near-orthogonal vectors have small pairwise inner products, so their squared sum norm is approximately the sum of their squared norms. If some contrast concepts are correlated (violating Assumption 1 for those pairs), their feature vectors share a common component that persists in $\bar{\mathbf{f}}_{-1}$ regardless of k, producing systematic bias. Stratified sampling of contrast concepts mitigates this.*

5 The SVD Estimator

An alternative is to look for the dominant direction across all difference vectors simultaneously via singular value decomposition. Stack *all* individual difference vectors (not pre-averaged) as rows:

$$V = \begin{pmatrix} (\mathbf{a}_{\text{pos}}^{(1,2)} - \mathbf{a}_{\text{neg}}^{(1,2)})^\top \\ \vdots \\ (\mathbf{a}_{\text{pos}}^{(n_2,2)} - \mathbf{a}_{\text{neg}}^{(n_2,2)})^\top \\ \vdots \\ (\mathbf{a}_{\text{pos}}^{(1,k)} - \mathbf{a}_{\text{neg}}^{(1,k)})^\top \\ \vdots \\ (\mathbf{a}_{\text{pos}}^{(n_k,k)} - \mathbf{a}_{\text{neg}}^{(n_k,k)})^\top \end{pmatrix} \in \mathbb{R}^{N \times d}, \tag{2}$$

where $N = \sum_{j=2}^{k} n_j$ is the total number of difference vectors. The idea is that $\mathbf{f}_1$ appears in every row as a common signal, while the reference features and per-sample noise vary across rows. The SVD should therefore pick up $\mathbf{f}_1$ as the leading direction.

Note that this formulation implicitly weights concept j proportionally to n_j (concepts with more prompt pairs contribute more rows). If equal concept weighting is desired, one can either balance n_j across concepts or pre-average within each concept before stacking, yielding a $(k-1) \times d$ matrix.

5.1 Analysis in the population limit

To characterize the SVD behavior, we analyze the idealized setting where per-sample noise $\boldsymbol{\eta}^{(i)}$ is negligible (absorbed into Assumption 3) and we work with one representative vector per concept contrast. This gives the $(k-1) \times d$ matrix:

$$V_{\text{ideal}} = \mathbf{1}\,\mathbf{f}_1^\top - F_{-1}, \tag{3}$$

where $\mathbf{1} \in \mathbb{R}^{k-1}$ is the all-ones vector and F_{-1} has rows $\mathbf{f}_2^\top, \ldots, \mathbf{f}_k^\top$. The first term is rank-1 with all its energy along $\mathbf{f}_1$; the second term spreads energy across the reference directions.

The Gram matrix is:

$$V_{\text{ideal}}^\top V_{\text{ideal}} = (k-1)\mathbf{f}_1\mathbf{f}_1^\top + \sum_{j=2}^{k} \mathbf{f}_j\mathbf{f}_j^\top - \mathbf{f}_1 S^\top - S\mathbf{f}_1^\top, \tag{4}$$

where $S = \sum_{j=2}^{k} \mathbf{f}_j$. The leading term gives an eigenvalue $\sim (k-1)\|\mathbf{f}_1\|^2$ along $\mathbf{f}_1$. The second contributes $\sim C^2$ along each $\mathbf{f}_j$. The cross terms $-\mathbf{f}_1 S^\top - S\mathbf{f}_1^\top$ have operator norm $O(C^2\sqrt{k-1})$—this is structural, arising because each row of V_{ideal} couples $\mathbf{f}_1$ to a different $\mathbf{f}_j$.

By the Davis–Kahan $\sin\theta$ theorem:

$$\sin\angle(\mathbf{u}_1, \mathbf{f}_1) \leq \frac{O(C^2\sqrt{k})}{(k-1)\|\mathbf{f}_1\|^2 - O(C^2\sqrt{k})} = O\left(\frac{1}{\sqrt{k}}\right). \tag{5}$$

5.2 Exact result for orthonormal features

When $\langle \mathbf{f}_i, \mathbf{f}_j \rangle = \delta_{ij}$ and $\|\mathbf{f}_i\| = 1$, the left Gram matrix is $V_{\text{ideal}}V_{\text{ideal}}^\top = \mathbf{1}\mathbf{1}^\top + I_{k-1}$, with eigenvalues k (eigenvector $\mathbf{1}/\sqrt{k-1}$) and 1 (multiplicity $k-2$). The leading right singular vector is:

$$\mathbf{u}_1 = \frac{(k-1)\mathbf{f}_1 - \sum_{j=2}^{k} \mathbf{f}_j}{\sqrt{k(k-1)}}, \tag{6}$$

giving $\sin\angle(\mathbf{u}_1, \mathbf{f}_1) = 1/\sqrt{k}$, confirming the rate is tight.

6 Equivalence of Mean and SVD

The mean estimator computes $V_{\text{ideal}}^\top \mathbf{1}$ (up to normalization). The SVD computes $V_{\text{ideal}}^\top \mathbf{u}_1^{(\text{left})}$, where $\mathbf{u}_1^{(\text{left})}$ is the leading left singular vector. Under our assumptions, the signal in V_{ideal} lives entirely in the all-ones direction of the row space: $\mathbf{f}_1$ contributes equally to every row. The SVD therefore finds $\mathbf{u}_1^{(\text{left})} \approx \mathbf{1}/\sqrt{k-1}$, and the two estimators agree to leading order.

The SVD does not extract more signal than the mean because the signal is rank-1 in the row space. The remaining singular components capture the spread of the reference features, which does not help recover $\mathbf{f}_1$.

When the full (non-averaged) matrix V is used, the equivalence holds approximately if concept contrasts have similar numbers of pairs. If n_j varies substantially, the SVD implicitly upweights concepts with more pairs, and the two estimators diverge for this reason alone.

7 When Mean and SVD Diverge: Diagnostics

Since the estimators differ exactly when $\mathbf{u}_1^{(\text{left})} \neq \mathbf{1}/\sqrt{k-1}$—that is, when the SVD implicitly *reweights* the contrastive pairs—their divergence signals that one or more assumptions are not holding. Inspecting the entries of $\mathbf{u}_1^{(\text{left})}$ can help identify which:

Observation on $\mathbf{u}_1^{(\text{left})}$	**Likely cause**	**Assumption**
Some entries near zero	Noisy or confounded pairs	A3
Clusters of similar weights	Redundant/correlated contrasts	A1
Weights proportional to row norms	Unequal feature magnitudes	A2
No clear pattern; $\sigma_2 \approx \sigma_1$	Target may be multi-dimensional	Setup

Low weight on a particular entry of $\mathbf{u}_1^{(\text{left})}$ is ambiguous on its own—it could indicate a noisy pair, a redundant pair, or a genuine assumption violation—but it flags which pairs deserve further scrutiny (e.g., checking row norms, pairwise correlations among rows, singular value spectrum).

Singular value spectrum. If σ_2/σ_1 is not small, the target concept may not be well-described by a single direction at the given layer and context. In this case, the leading singular *subspace* (top-r right singular vectors) may be more appropriate.

8 All Pairwise Contrasts

If we also collect difference vectors for pairs not involving $\mathbf{f}_1$ (i.e., $\mathbf{f}_i - \mathbf{f}_j$ for all $i \neq j$), we obtain $\binom{k}{2}$ sets of averaged difference vectors in k unknowns (the feature vectors, up to a global translation). The incidence matrix of the complete graph on k vertices has rank $k-1$, so fixing the translation ambiguity (e.g., $\sum_i \mathbf{f}_i = \mathbf{0}$) makes the system overdetermined.

The least-squares solution recovers each feature individually: $\hat{\mathbf{f}}_i = \frac{1}{k}\sum_{j \neq i} \mathbf{v}_{ij}$. Under all three assumptions, the error is $O(\epsilon C)$—the near-orthogonality residual—rather than the $O(C/\sqrt{k})$ from star-topology contrasts. The cost is quadratic in the number of concept-pair datasets that must be constructed, which may be prohibitive.

This observation clarifies that the $1/\sqrt{k}$ rate is a consequence of the star-shaped observation topology (all contrasts through $\mathbf{f}_1$), not a fundamental limit of the problem.

9 Discussion

Relation to standard steering. In standard CAA [1, 2], one steers along $\mathbf{f}_1 - \mathbf{f}_j$: the axis of variation between two concepts. This is effective precisely because it captures both features—it tells the model to move toward concept 1 and away from concept j. Our method does not replace this; rather, it addresses a different goal. By averaging across multiple reference concepts, we obtain a direction closer to $\mathbf{f}_1$ itself, which is useful when one wants a direction that generalizes to held-out concepts or when one wants to validate that a hypothesized feature exists as a stable linear direction.

Computational equivalence with multi-pair steering. If one wants to steer from each of $\mathbf{f}_2, \ldots, \mathbf{f}_k$ to $\mathbf{f}_1$, one already needs to construct $k-1$ contrastive datasets—the same data our method requires. Computing $k-1$ individual steering vectors and then deploying the appropriate one per context is one approach; averaging them to obtain a single direction closer to $\mathbf{f}_1$ is another. The latter has the advantage of generalizing to steering from concepts $\mathbf{f}_l$ not seen during construction, since it depends less on any particular reference.

Validation of feature hypotheses. Perhaps the most immediate use is as a consistency check. If one hypothesizes that "English" is a linear feature at layer ℓ, constructing contrasts against several other languages and checking that the method produces a stable direction (low σ_2/σ_1, consistent $\mathbf{u}_1^{(\text{left})}$ weights) provides evidence for this hypothesis at the given layer and context. Failure modes—high σ_2/σ_1, divergence between mean and SVD, instability across subsets of contrasts—are informative about the structure of the representation. Note that this requires already having an inkling of what the features are in order to construct the contrastive datasets; the method validates a hypothesis rather than discovering features *de novo*.

Representation engineering. Representation engineering [3] applies PCA to contrastive activations, finding directions of maximum variance across prompts within a single concept contrast. Our SVD operates across concept contrasts rather than across prompts, and the analysis shows when and why the leading singular vector approximates the target feature.

Sparse autoencoders. SAEs [4, 5] learn overcomplete dictionaries of feature directions from activations. Multi-contrastive projection is complementary: it targets a specific concept using external structure (contrastive pairs), while SAEs discover features unsupervised. The two could be combined—e.g., using multi-contrastive directions to validate or seed SAE features for specific target concepts.

Semantic similarity. If some contrast concepts are semantically close (e.g., several Romance languages), Assumption 1 may not hold for those pairs. Their shared component would persist in the average regardless of k, producing systematic bias. Whether this is a serious issue in practice depends on the model and concepts involved; we flag it as a consideration rather than a demonstrated failure mode. Stratified sampling of contrast concepts (ensuring diversity across semantic categories) is a natural mitigation.

References

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